Mathematics: analysis and approaches
Standard level
Paper 2 — markscheme
2.1–2.5 Functions

1 November 2026

W4ss Academy practice set 1 | **Topic** 2.1–2.5 Functions | **Ref.** W4SS–MAA–T2–P2–01

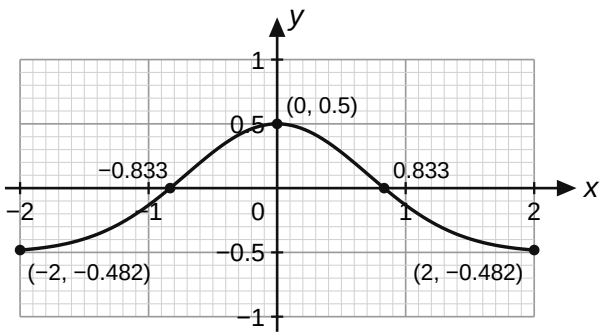
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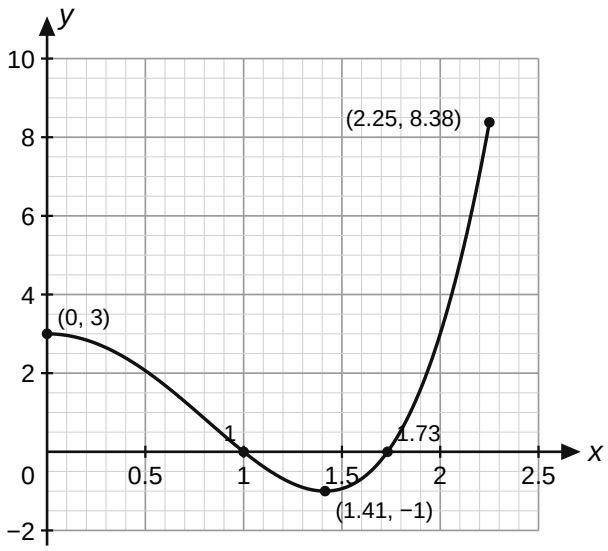
Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- The question export used for this paper has no official markscheme: these solutions were written by W4ss Academy in the IB style. Accept any correct equivalent method.
- Total: **[25 marks]**.

Section A

Question	Marks	Markscheme
1. (a)	**[2]**	$x = -0.833$, $x = 0.833$ $(\pm 0.832554\ldots$, i.e. $\pm\sqrt{\ln 2})$ ***A1A1***
1. (b)	**[3]**	 correct shape: symmetrical about the y-axis, with a single maximum on the y-axis ***A1*** maximum at (0, 0.5) and x-intercepts at about ± 0.833 ***A1*** end points at about (−2, −0.482) and (2, −0.482) ***A1***
2. (a)(i)	**[2]**	$x + \frac{12}{x^2} = 0$, or the x-intercept of the graph on the GDC ***(M1)*** $x = -2.29$ $(-2.28942\ldots)$ ***A1*** **Note:** Accept (−2.29, 0).
2. (a)(ii)	**[2]**	(2.88, 4.33) ((2.88449…, 4.32674…)) ***A1A1*** **Note:** A1 for each coordinate. Exact values: $(2\sqrt[3]{3}, 3\sqrt[3]{3})$.
2. (b)	**[2]**	$x + \frac{12}{x^2} = 3 - x$, or the graphs of f and g on the GDC ***(M1)*** $x = -1.43$ $(-1.43080\ldots)$ ***A1***
3. (a)	**[2]**	minimum value of f is −5 (at $x = 1$), e.g. $f(x) = 6(x-1)^2 - 5$ or from the GDC ***(A1)*** $f(x) \geq -5$ ***A1***
3. (b)	**[4]**	$(g \circ f)(x) = -(6x^2 - 12x + 1) + c = -6x^2 + 12x - 1 + c$ ***(A1)*** recognizing that the maximum of $g \circ f$ is at $x = 1$ (or that $c \leq f(x)$ for every x) ***(M1)*** maximum value of $(g \circ f)(x) = 5 + c$ ***(A1)*** $5 + c \leq 0 \Rightarrow c \leq -5$ ***A1***

4. (a)	**[2]**	$(f \circ g)(x) = f(x^2 - 2) = (x^2 - 2)^2 - 1$ ***A1*** $= x^4 - 4x^2 + 4 - 1$ ***A1*** $= x^4 - 4x^2 + 3$ ***AG***
4. (b)	**[3]**	 correct shape: decreasing to a single minimum, then increasing ***A1*** *y*-intercept (0, 3), *x*-intercepts at 1 and about 1.73, minimum at about (1.41, −1) ***A1*** curve stops at $x = 2.25$, at about (2.25, 8.38) ***A1***
4. (c)	**[3]**	recognizing the minimum value −1 and $(f \circ g)(0) = 3$, e.g. horizontal lines on the graph ***(M1)*** $-1 < k \leq 3$ ***A1A1*** **Note:** A1 for $-1 < k$, A1 for $k \leq 3$. For $k = 3$ the two solutions are $x = 0$ and $x = 2$.

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Answers written on this page
will not be marked.